WASP-10b

R-1352

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_211004` · created 2026-07-31T22:47:38+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459491.7778 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.034
Transit depth (Rp/Rs)^2	1.67 +/- 0.88 [%]
Orbital Inclination (inc)	87.2 +/- 1.46
Airmass coefficient 1 (a1)	0.829 +/- 0.016
Airmass coefficient 2 (a2)	0.164 +/- 0.031
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	#2 - [385, 316]
Optimal Aperture	4.03
Optimal Annulus	12.57
Transit Duration (day)	0.075 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.129 ± 0.034 vs 0.1592 (deviation 0.72σ)
Duration fit vs published	0.075 d vs 0.0946 d (deviation 0.8σ)
Mid-time O–C	-12.5 min
Depth SNR	3.1
Residual scatter	1.89 %

Light curve

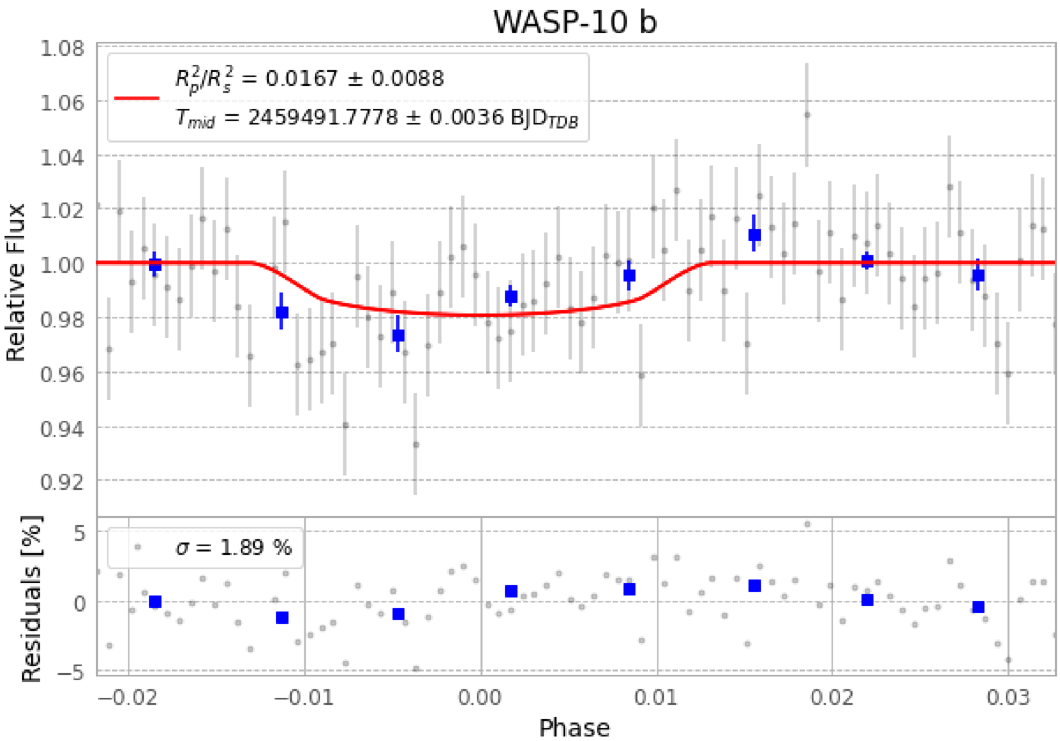

Figure 1 — FinalLightCurve_WASP-10 b_2021-10-03.png (SHA-256 6b354d72fa7d4bbd…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [400, 185], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 eabcf45e3d19cbd0…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

86 FITS frames (56 MB), `WASP-10211004044212.FITS` → `WASP-10211004085712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-211004.log (674d45a23c7d…), FinalLightCurve_WASP-10 b_2021-10-03.png (6b354d72fa7d…), FinalLightCurve_WASP-10 b_2021-10-03.csv (51512b1b6733…)

Compute resources

Wall time 295.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)